Euler Method

Taylor series method is the fundamental numerical method for the solution of the initial value problem

Expanding $y(x)$ in Taylor series about any point x_i, with the Lagrange form of remainder, we obtain

$$y(x) = y(x_i) + (x - x_i)\, y'(x_i) + \frac{1}{2!}(x - x_i)^2 y''(x_i) + \ldots + \frac{1}{p!}(x - x_i)^p y^{(p)}(x_i)$$

$$+ \frac{1}{(p+1)!}(x - x_i)^{p+1} y^{(p+1)}(x_i + \theta h) \qquad (1)$$

where $0 < \theta < 1$, $x \in [x_0, b]$ and b is the point up to which the solution is required.

We denote the numerical solution and the exact solution at x_i by y_i and $y(x_i)$ respectively.

Now, consider the interval $[x_i, x_{i+1}]$. The length of the interval is $h = x_{i+1} - x_i$.

Substituting $x = x_{i+1}$ in (1) , we obtain

$$y(x_{i+1}) = y(x_i) + h\, y'(x_i) + \frac{h^2}{2!} y''(x_i) + \ldots + \frac{h^p}{p!} y^{(p)}(x_i) + \frac{h^{p+1}}{(p+1)!} y^{(p+1)}(x_i + \theta h).$$

Neglecting the error term, we obtain the *Taylor series method* as

$$y_{i+1} = y_i + hy_i' + \frac{h^2}{2!} y_i'' + \ldots + \frac{h^p}{p!} y_i^{(p)}. \tag{2}$$

Note that Taylor series method is an explicit single step method.

the truncation error of the method is given by

$$T_{i+1} = \frac{h^{p+1}}{(p+1)!} y^{(p+1)} (x_i + \theta h). \tag{3}$$

Using the definition of the order , we say that the Taylor series method (2) is of order p.

For $p = 1$, we obtain the first order Taylor series method as

$$y_{i+1} = y_i + hy_i' = y_i + hf(x_i, y_i). \tag{4}$$

This method is also called the *Euler method*. The truncation error of the Euler's method is

$$\text{T.E.} = \frac{h^2}{2!} y_i''(x_i + \theta h). \tag{5}$$

Example *Solve the initial value problem* $yy' = x$, $y(0) = 1$, *using the Euler method in* $0 \le x \le 0.8$, *with* $h = 0.2$ *and* $h = 0.1$. *Compare the results with the exact solution at* $x = 0.8$.

Solution We have $y' = f(x, y) = (x/y)$.

Euler method gives $y_{i+1} = y_i + h\, f(x_i, y_i) = y_i + \dfrac{hx_i}{y_i}$.

Initial condition gives $x_0 = 0, y_0 = 1$.

When $h = 0.2$, we get $y_{i+1} = y_i + \dfrac{0.2\, x_i}{y_i}$.

We have the following results.

$$y(x_1) = y(0.2) \approx y_1 = y_0 + \frac{0.2\, x_0}{y_0} = 1.0.$$

$$y(x_2) = y(0.4) \approx y_2 = y_1 + \frac{0.2\, x_1}{y_1} = 1.0 + \frac{0.2(0.2)}{1.0} = 1.04.$$

$$y(x_3) = y(0.6) \approx y_3 = y_2 + \frac{0.2\,x_2}{y_2} = 1.04 + \frac{0.2(0.4)}{1.04} = 1.11692$$

$$y(x_4) = y(0.8) \approx y_4 = y_3 + \frac{0.2\,x_3}{y_3} = 1.11692 + \frac{0.2(0.6)}{1.11692} = 1.22436.$$

When $h = 0.1$, we get $y_{i+1} = y_i + \dfrac{0.1\,x_i}{y_i}$.

We have the following results.

$$y(x_1) = y(0.1) \approx y_1 = y_0 + \frac{0.1x_0}{y_0} = 1.0.$$

$$y(x_2) = y(0.2) \approx y_2 = y_1 + \frac{0.1x_1}{y_1} = 1.0 + \frac{0.1(0.1)}{1.0} = 1.01.$$

$$y(x_3) = y(0.3) \approx y_3 = y_2 + \frac{0.1x_2}{y_2} = 1.01 + \frac{0.1(0.2)}{1.01} = 1.02980.$$

$$y(x_4) = y(0.4) \approx y_4 = y_3 + \frac{0.1x_3}{y_3} = 1.0298 + \frac{0.1(0.3)}{1.0298} = 1.05893.$$

$$y(x_5) = y(0.5) \approx y_5 = y_4 + \frac{0.1x_4}{y_4} = 1.05893 + \frac{0.1(0.4)}{1.05893} = 1.09670.$$

$$y(x_6) = y(0.6) \approx y_6 = y_5 + \frac{0.1x_5}{y_5} = 1.0967 + \frac{0.1(0.5)}{1.0967} = 1.14229.$$

$$y(x_7) = y(0.7) \approx y_7 = y_6 + \frac{0.1x_6}{y_6} = 1.14229 + \frac{0.1(0.6)}{1.14229} = 1.19482.$$

$$y(x_8) = y(0.8) \approx y_8 = y_7 + \frac{0.1x_7}{y_7} = 1.19482 + \frac{0.1(0.7)}{1.19482} = 1.25341.$$

The exact solution is $y = \sqrt{x^2 + 1}$.

At $x = 0.8$, the exact value is $y(0.8) = \sqrt{1.64} = 1.28062$.

The magnitudes of the errors in the solutions are the following:

$$h = 0.2: \ | \ 1.28062 - 1.22436 \ | = 0.05626.$$

$$h = 0.1: \ | \ 1.28062 - 1.25341 \ | = 0.02721.$$

Example *Consider the initial value problem* $y' = x(y + 1)$, $y(0) = 1$. *Compute* $y(0.2)$ *with* $h = 0.1$ *using (i) Euler method (ii) Taylor series method of order two*

Solution We have $f(x, y) = x(y + 1)$, $x_0 = 0$, $y_0 = 1$.

(*i*) Euler's method: $y_{i+1} = y_i + h\, f(x_i, y_i) = y_i + 0.1[x_i\,(y_i + 1)]$.

With $x_0 = 0$, $y_0 = 1$, we get

$$y(0.1) \approx y_1 = y_0 + 0.1[x_0\,(y_0 + 1)] = 1 + 0.1[0] = 1.0.$$

With $x_1 = 0.1$, $y_1 = 1.0$, we get

$$y(0.2) \approx y_2 = y_1 + 0.1\,[x_1(y_1 + 1)]$$
$$= 1.0 + 0.1[(0.1)(2)] = 1.02.$$

(*ii*) Taylor series second order method.

$$y_{i+1} = y_i + h\, y_i' + \frac{h^2}{2!}\, y_i'' = y_i + 0.1\, y_i' + 0.005 y_i''.$$

We have $y'' = xy' + y + 1.$

With $x_0 = 0, y_0 = 1$, we get

$$y_0' = 0, y_0'' = x_0 y_0' + y_0 + 1 = 0 + 1 + 1 = 2.$$

$$y(0.1) \approx y_1 = y_0 + 0.1 y_0' + 0.005 y_0''$$

$$= 1 + 0 + 0.005\ [\ 2] = 1.01.$$

With $x_1 = 0.1, y_1 = 1.01$, we get

$$y_1' = 0.1(1.01 + 1) = 0.201.$$

$$y_1'' = x_1 y_1' + y_1 + 1 = (0.1)(0.201) + 1.01 + 1 = 2.0301.$$

$$y(0.2) \approx y_2 = y_1 + 0.1\, y_1' + 0.005 y_1'$$

$$= 1.01 + 0.1\ (0.201) + 0.005(2.0301)] = 1.04025.$$